

Module 12 — Question Pool

OMSCS 6250 Computer Networks

Lesson 12: Applications — CDNs and Overlay Networks

Introduction to Content Distribution Networks

Q1. [MCQ] Why CDN architecture was designed

Why was the CDN architecture invented? Which is the BEST single answer?

- **A.** Serving popular content from many geographically distributed locations reduces latency to users and avoids repeatedly sending the same bytes across long-haul links.
- **B.** CDNs were invented to replace BGP as the inter-AS routing protocol for popular content, letting providers advertise their own paths and bypass Tier-1 ISP route selection.
- **C.** CDNs were invented to replace TCP with a lower-overhead transport, since TCP's per-window ACKs make the server-to-user distance the dominant bottleneck.
- **D.** CDNs were invented so a provider could serve all users from one upgraded 'first-mile' data center, removing the need for geographically distributed servers.

Correct answer: **A**

Why: Replicate popular content geographically => lower RTT + no repeat long-haul transfers. CDNs solve the WAN bottleneck: instead of every user pulling from a distant origin, content lives near each user.

Q2. [TF] What a CDN is

A CDN is a network of geographically distributed servers that hold REPLICATED copies of content and steer users to a server cluster that can best serve their request.

- True
- False

Correct answer: **True**

Why: CDN = distributed replicas + smart steering. Many edge clusters host the same content; the CDN steers each user to a cluster that's geographically close, lightly loaded, and on a healthy network path.

Q3. [MCQ] Single-datacenter drawbacks

Which is NOT a drawback of serving all content from a single, massive data center?

- **A.** Large geographic distance to globally distributed users.
- **B.** The simplicity of managing all content in a single location.

- C. Repeated transmission of the same popular content across the same long-haul links.
- D. Single point of failure for outages or natural disasters.

Correct answer: **B**

Why: The three drawbacks in the lesson are distance, repeated long-haul transfer of identical content, and a single point of failure. Managing everything in one place is actually the ADVANTAGE of the single-DC approach - the lesson calls it 'a simple design' - so it is not a drawback.

Content Delivery Challenges

Q4. [MCQ] The middle-mile bottleneck

Among the six content-delivery challenges, why does the 'middle mile' (the peering points between networks) tend to stay congested?

- A. The middle mile is the coax 'last-mile' link into each subscriber's home, which is inherently low-bandwidth.
- B. Peering points are required to run only IPv6, which most core routers cannot forward, so packets are dropped.
- C. There is business incentive to upgrade the first mile and the last mile, but little incentive to upgrade the peering points between networks (no revenue there) — so they become bottlenecks with packet loss and higher latency.
- D. Tier-1 ISPs deliberately throttle peering points to push content providers onto paid transit.

Correct answer: **C**

Why: Middle-mile = peering points with no revenue to justify upgrades, so they stay congested (packet loss, latency). First mile (hosts) and last mile (users) get the investment.

Q5. [MCQ] The scalability challenge

The lesson lists 'scalability' as a challenge for Internet applications. What makes it hard?

- A. TCP cannot open more than one connection per server, capping total throughput.
- B. BGP forbids advertising more than one route per prefix, limiting the number of servers.
- C. DNS TTLs must be set to zero during spikes, which overwhelms resolvers.
- D. Demand can spike unpredictably (a video going viral, Black Friday), yet scaling up infrastructure is expensive, slow, and hard to forecast.

Correct answer: **D**

Why: Scalability = matching resources to demand that spikes unpredictably; scaling infrastructure is costly, slow, and hard to forecast.

CDNs and the Internet Ecosystem

Q6. [MCQ] Flattening the hierarchy

Why does CDN deployment FLATTEN the Internet hierarchy, as discussed in this lesson?

- **A.** CDNs operate at Layer 2, switching frames within one access network, so they leave the inter-AS routing hierarchy intact.
- **B.** CDNs replace BGP with a custom routing protocol.
- **C.** CDNs are forbidden from peering at any IXP, which keeps the hierarchy intact instead.
- **D.** Edge caches let large amounts of traffic be exchanged locally with access networks — bypassing the climb up to Tier-1 ISPs for distant origins.

Correct answer: **D**

Why: Edge caches let bulk traffic stay LOCAL => no climb to Tier-1s. Big content (Netflix, YouTube) is served from caches at IXPs or inside access networks, so most bytes never traverse the upper hierarchy.

CDNs Server Placement Approaches

Q7. [MCQ] Enter Deep (Akamai)

What does the 'Enter Deep' placement philosophy look like, and which CDN exemplifies it?

- **A.** Many small clusters deployed deep inside access networks, close to end users — Akamai is the canonical example.
- **B.** A few VERY LARGE clusters concentrated at key IXPs only.
- **C.** A single mega-cluster in a central data center, with no edge presence.
- **D.** Mobile clusters that physically relocate based on user demand.

Correct answer: **A**

Why: Enter Deep = many SMALL clusters deep in access nets (Akamai). Push servers as close to eyeballs as possible — even into ISP networks — maximising last-mile latency wins at the cost of operational complexity.

Q8. [MCQ] Bring Home

What does the 'Bring Home' placement philosophy look like?

- **A.** Many small clusters deep inside access networks, close to end users.
- **B.** fewer, larger clusters at a small number of key points (often major IXPs) — content travels a bit farther from server to user but is easier to operate.
- **C.** A single hosted cluster placed inside each end-user's home router, so content is served from the customer premises and rarely leaves the last mile.
- **D.** No clusters; the origin server handles every request.

Correct answer: **B**

Why: Bring Home = FEWER, LARGER clusters at IXPs. Limelight-style: concentrate capacity at major peering points so content travels a bit farther but the deployment is simpler to run.

Q9. [TF] Placement is a spectrum

'Enter Deep' and 'Bring Home' are mutually exclusive strategies: a CDN must commit to one, and real deployments never combine deep edge nodes with large clusters at IXPs.

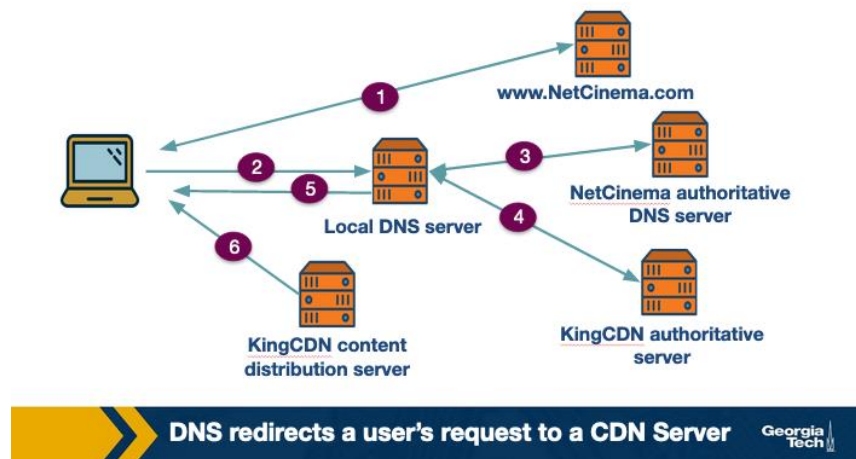
- True
- False

Correct answer: **False**

Why: False. The lesson says CDNs employ a HYBRID — e.g., Google runs mega-datacenters, ~50 clusters at IXPs, AND hundreds of clusters deep in access ISPs. Enter Deep and Bring Home are endpoints of a spectrum that real CDNs combine.

How a CDN Operates

Q10. [MCQ] First step: the DNS query



Reference Figure: How a CDN operates — DNS-mediated redirection (Module 12)

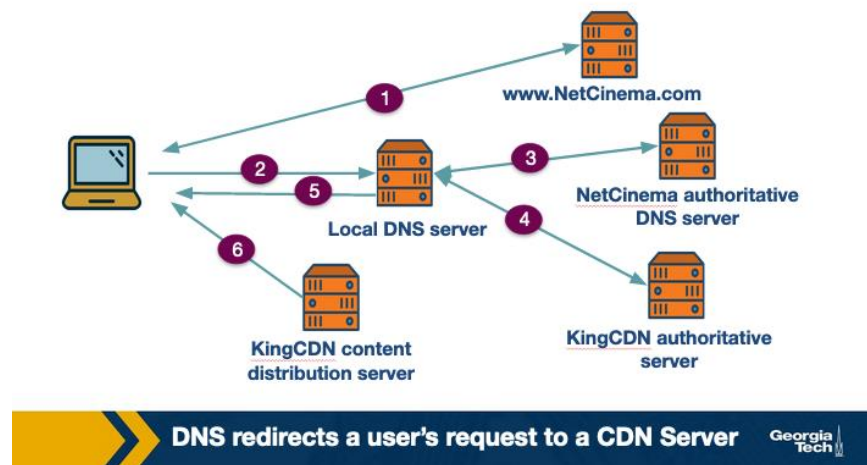
When the user clicks <http://video.examplemovies.com/R2D2C3PO37>, what happens FIRST?

- A. The CDN's name server immediately pushes an IP address to the client over UDP.
- B. The client opens a TCP connection directly to the CDN edge server.
- C. The host issues a DNS query for 'video.examplemovies.com' to its local DNS resolver (LDNS).
- D. ExampleMovies streams the video directly from its origin server with no DNS step.

Correct answer: **C**

Why: Step 1: client -> LDNS for the content hostname. DNS resolution is the first network step; everything CDN-related is bootstrapped through name resolution before any TCP handshake.

Q11. [MCQ] Handoff to the CDN's DNS



Reference Figure: How a CDN operates — DNS chain through ExampleCDN (Module 12)

ExampleMovies's authoritative DNS responds to the query for 'video.examplemovies.com' by returning:

- A. An anycast prefix advertisement carried in DNS.
- B. The IP of ExampleMovies's origin server directly, because its authoritative DNS keeps a static A-record and does not delegate the name.
- C. A 3xx HTTP redirect to a CDN URL, bypassing DNS.
- D. A hostname in ExampleCDN's domain (e.g., a1130.examplecdn.com) — the CDN's DNS then resolves to the chosen edge server.

Correct answer: **D**

Why: Origin's DNS returns CNAME into CDN's domain => CDN's DNS picks the edge. Content provider's DNS delegates to the CDN's name service, which makes the actual server-selection decision based on the client's LDNS.

Q12. [TF] DNS as the steering hook

CDNs intercept a user's content request and steer it to an edge server WITHOUT involving DNS — the browser is hard-coded with each edge server's IP address.

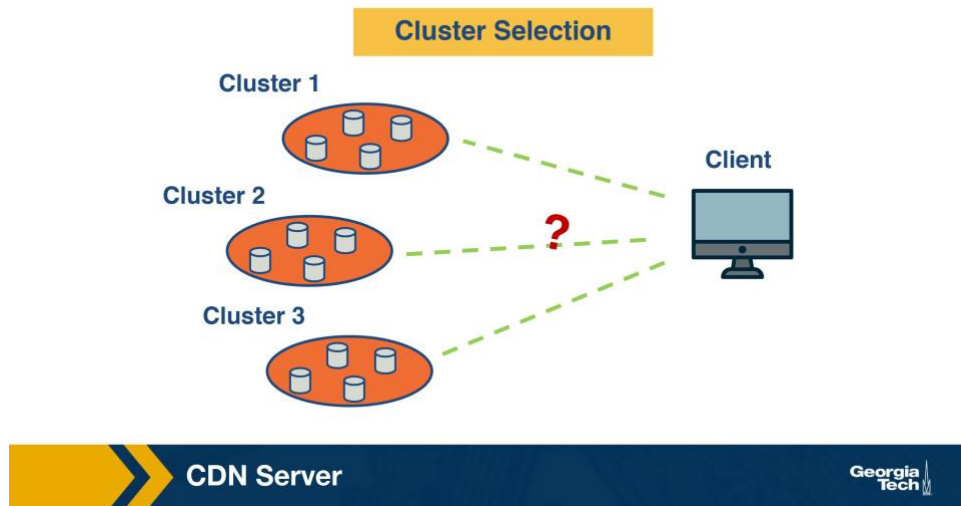
- True
- False

Correct answer: **False**

Why: False. DNS-based redirection IS the main steering mechanism: every request goes through DNS first, giving the CDN a chance to pick the best edge server before any TCP is opened. Browsers are not hard-coded with edge IPs.

CDN Server Selection

Q13. [MCQ] Two selection granularities



Reference Figure: CDN server selection — multi-step pipeline (Module 12)

From the server-selection figure, server selection happens at TWO granularities. Which pair is correct?

- A. cluster selection (which geographic site to serve from) and server selection within that cluster (which specific server).
- B. Cluster selection by the user and server selection by the CDN; nothing else.
- C. Server selection per packet at the IP layer and cluster selection per Ethernet frame at L2.
- D. Cluster selection by the origin server and server selection by the IXP route server.

Correct answer: **A**

Why: Two granularities: cluster (which site) + server (within cluster). First decide which geographic cluster handles the request; then within that cluster, pick a specific server (e.g., consistent hashing).

Cluster Selection Strategies

Q14. [MCQ] Geographic cluster selection

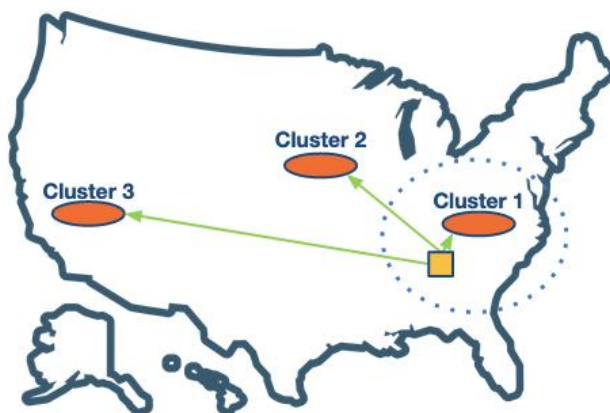


Figure: Simple geographic cluster selection — Atlanta → Charlotte (Module 12)

Consider a CDN with clusters in Charlotte, Boston, and Los Angeles, under simple geographic selection, where is a user at Georgia Tech (Atlanta) served from?

- A. Boston — closest by IP-anycast metric in this figure.
- B. Charlotte — the geographically closest cluster.
- C. Los Angeles — most populated CDN cluster.
- D. The origin server — geographic selection only applies to peer-served content.

Correct answer: **B**

Why: Atlanta → Charlotte (closest cluster geographically). Simple geographic selection uses distance only; Charlotte is the closest among Charlotte/Boston/LA to Georgia Tech.

Q15. [TF] Closest is not always best

Picking the closest cluster 'as the crow flies' may STILL yield a suboptimal end-to-end path because of BGP routing inefficiencies or transient link congestion.

- True
- False

Correct answer: **True**

Why: Geographic closeness != best end-to-end path. BGP routing inefficiencies, congested links, or transit relationships can make a 'farther' cluster actually faster — pure geography misses these effects.

Q16. [MCQ] Active vs passive measurement

Why is using ACTIVE measurements (e.g., having the LDNS ping candidate clusters) impractical for real-time cluster selection?

- A. Active probing requires BGP to be temporarily disabled on the client's AS.
- B. ICMP ping is blocked by every CDN, so RTT can never be measured actively.
- C. Active measurements are limited to once per day by DNS TTL rules.
- D. Most LDNS are not equipped to perform active probing, and doing so would generate a lot of extra traffic.

Correct answer: **D**

Why: Active measurement (LDNS pinging clusters) is impractical: most LDNS can't do it and it floods the network with probes. CDNs instead use passive measurements from existing sessions.

Policy for Server Selection

Q17. [MCQ] Cache miss: fetch from origin

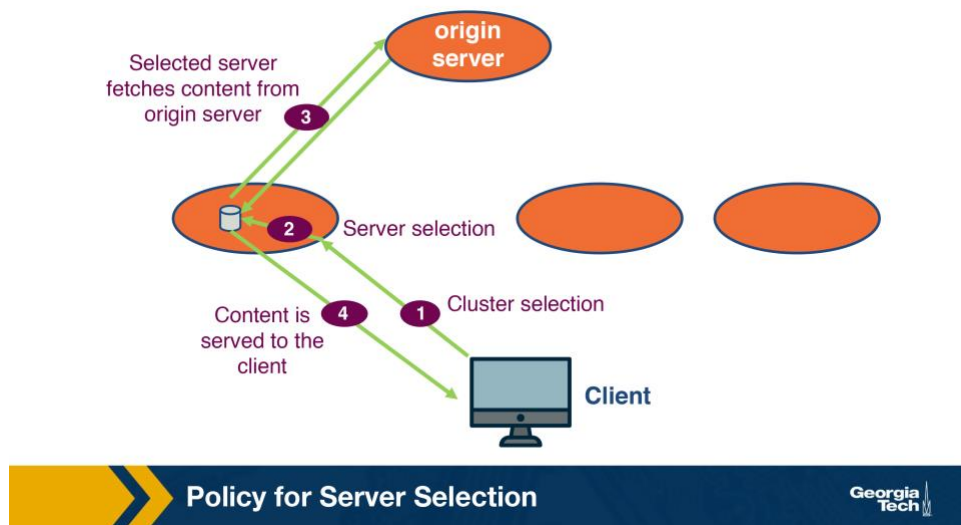


Figure: Within-cluster server selection — lazy fetch from origin and cache (Module 12)

From the policy figure, scenario: a request arrives at a cluster but the selected server does NOT have the content cached. What happens next?

- **A.** The server fetches the content from an origin server (or peer), then serves and caches it for future requests.
- **B.** The request is dropped and the client retries automatically.
- **C.** DNS is re-queried for a new cluster across the country.
- **D.** The client is redirected via IP anycast to a different cluster that is expected to already hold the requested object.

Correct answer: **A**

Why: Cache miss => fetch from origin/peer, then cache. The first request pays the origin-fetch latency; subsequent requests at the same cluster hit the cache and are fast.

Q18. [TF] Lazy-fetch trade-off

Lazy fetching on cache miss means the FIRST request for a piece of content may be slower (origin fetch latency), but subsequent requests at the same cluster are fast.

- True
- False

Correct answer: **True**

Why: First request slow (origin fetch); later ones fast. The trade-off is on-demand caching: you don't pre-populate every server with every object, you fetch on miss.

Consistent Hashing

Q19. [MCQ] Mapping onto the hash ring

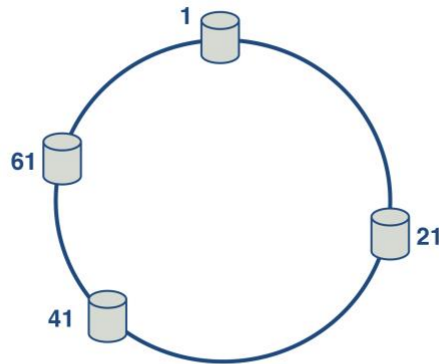


Figure: Consistent hashing — servers and objects on a ring (Module 12)

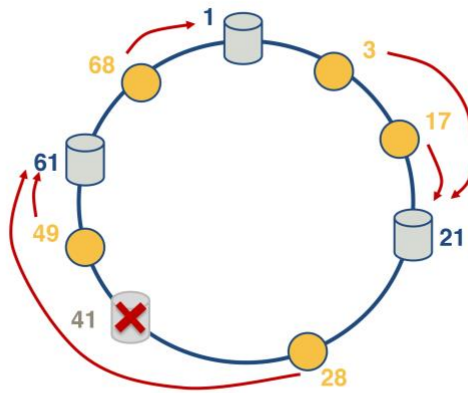
From the consistent-hashing figure, how are servers and content objects mapped to slots?

- A. Servers are mapped by hashing IPs while objects are hashed by URL into a separate offset table, so the two ID spaces stay independent.
- B. both servers and objects are mapped to the same ID space on a circle; an object is assigned to its successor server (the next server clockwise on the ring).
- C. Servers are mapped to slots, but objects are hashed into a separate offset table.
- D. Both are mapped randomly; lookup is by linear scan over the ring.

Correct answer: **B**

Why: Both servers AND objects on the SAME ring; object -> successor server. Hash both into the same key space; each object is served by the next server clockwise from its hash position.

Q20. [MCQ] When a server leaves



Consistent Hashing

Georgia Tech

Figure: Consistent hashing — server leaves the ring (Module 12)

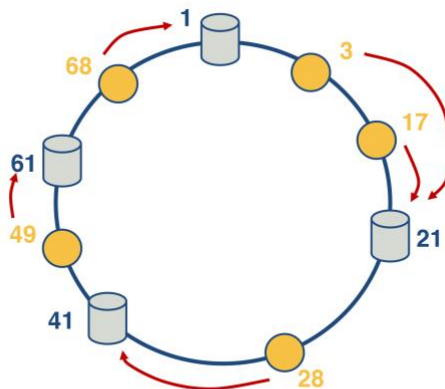
Scenario: in the figure, server #41 LEAVES the ring (fails or is decommissioned). What happens to objects it was serving?

- A. Objects are returned to the origin and removed from every cache.
- B. ALL objects in the ring are rehashed across the remaining servers from scratch.
- C. only those objects are remapped to the next (successor) server on the ring — most objects are unaffected.
- D. The hash function is recomputed for every server, producing a new mapping.

Correct answer: C

Why: Server leaves => only ITS objects move to successor. Beautiful property of consistent hashing: $\sim 1/N$ objects move when a server is added or removed, not all $N \times K$ objects.

Q21. [TF] Minimal remapping



Consistent Hashing

Georgia Tech

Figure: Consistent hashing — robustness to server additions/removals (Module 12)

Consistent hashing is designed so that adding or removing a server moves only a SMALL fraction of the objects between servers, rather than remapping nearly all of them.

- True
- False

Correct answer: **True**

Why: Consistent hashing moves only a small fraction of objects on join/leave - the lesson: 'relatively little movement of these content IDs when nodes join and leave.' Classic hash-mod-N would re-map nearly every object when N changes; consistent hashing limits the disruption to the neighborhood of the server that joined or left.

Network Protocols Used for Cluster/Server Selections

Q22. [MCQ] Steering protocols (not OSPF)

Which protocol is NOT one of the network mechanisms a CDN uses for cluster/server selection?

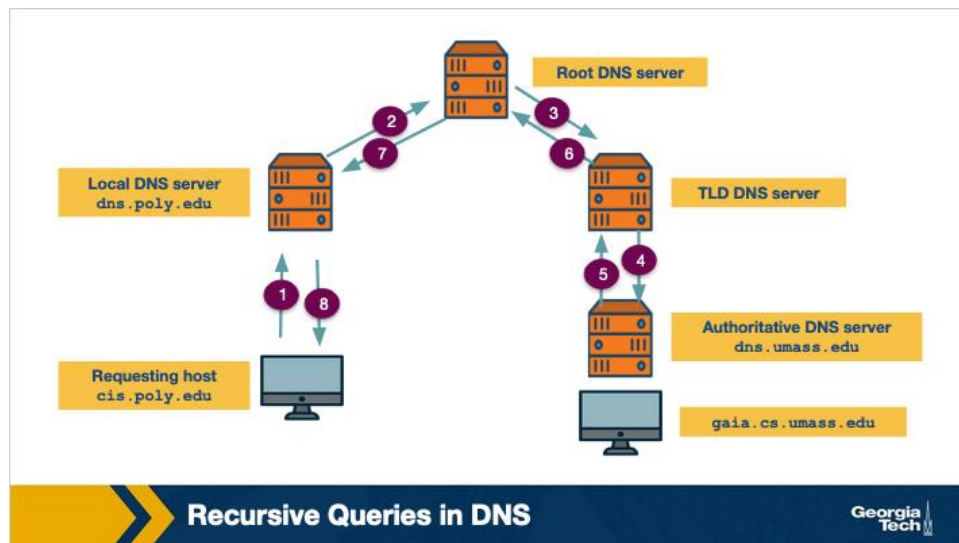
- A. IP anycast.
- B. DNS.
- C. HTTP redirection.
- D. OSPF.

Correct answer: **D**

Why: OSPF is intra-domain routing, not a CDN steering protocol. CDN steering uses DNS, IP anycast, and HTTP redirects; OSPF runs inside ISPs to compute internal paths.

Server Selection Strategies: The DNS Protocol

Q23. [MCQ] Recursive vs iterative queries



Reference Figure: DNS resolution — recursive then iterative (Module 12)

In the reference figure, a host requests `gaia.cs.umass.edu` via `dns.poly.edu`. Which type of query is typically used between the HOST and its LDNS?

- A. Recursive — the host asks the LDNS to resolve and return the final answer.
- B. Iterative — the host walks the DNS hierarchy step by step.
- C. Anycast — the host hashes the name to an anycast IP directly.
- D. HTTP redirect — DNS is bypassed at the client side.

Correct answer: **A**

Why: Host -> LDNS is **RECURSIVE**: LDNS returns the final answer. The host asks 'tell me the answer'; the LDNS does the iterative walk (root -> TLD -> authoritative) on the host's behalf.

Q24. [MCQ] Closest-to-LDNS limitation

A key **LIMITATION** of DNS-based CDN steering, when it tries to pick the 'closest' cluster, is that:

- A. DNS forces use of recursive queries only, which add many extra hops compared to iterative.
- B. The CDN's authoritative name server sees only the LDNS, not the client — so 'closest' really means closest to the LDNS, which may differ from the client's actual location.
- C. DNS bypasses the client's local resolver and queries the authoritative server directly, so every page load pays a full recursive walk from the root.
- D. DNS messages are too large for UDP, forcing TCP fallback on every lookup.

Correct answer: **B**

Why: Authoritative sees LDNS, not the client => 'closest to LDNS' may not equal closest to user. If a client uses a far-away DNS resolver (e.g., 8.8.8.8), the CDN picks a server near 8.8.8.8 rather than near the client.

Q25. [TF] DNS granularity limits

Because a DNS response can be recomputed on every single request, DNS-based redirection reacts to second-by-second congestion changes just as fast as an in-path mechanism.

- True
- False

Correct answer: **False**

Why: False. DNS-based steering works at the granularity of (LDNS-based) regional locality and is cached for the record's TTL, so it CANNOT react to second-by-second congestion; reacting that fast needs a different mechanism (e.g., HTTP redirect or anycast withdrawal).

Q26. [MCQ] DNS hierarchy order

To resolve amazon.com from scratch, in what order does the query traverse the DNS server hierarchy?

- A. Authoritative server -> root server -> a .com TLD server.
- B. Root server -> a .com TLD server -> the authoritative server for amazon.com.
- C. Local DNS server -> root server -> local DNS server, with no TLD or authoritative server involved.
- D. TLD server -> root server -> authoritative server.

Correct answer: **B**

Why: Root returns the TLD (.com) server; the TLD returns the authoritative server for amazon.com; the authoritative server returns the final A record.

More on DNS: Making Responses Faster with Caching

Q27. [MCQ] DNS caching and TTL

How does DNS CACHING speed up resolutions, and what does TTL control?

- A. TTL controls how often the DNS response's DNSSEC signature and encryption keys are rotated, not its cacheability, so resolvers still re-query the authoritative server each time.
- B. Only the client's browser caches the answer; the local DNS resolver never stores records, so every new query still walks the full root -> TLD -> authoritative hierarchy.
- C. Resolvers and clients cache resolved A-records for the duration of the record's TTL — repeated lookups within the TTL are served from cache without re-walking the hierarchy.
- D. TTL controls the size of the cache only, not the duration each entry is held.

Correct answer: **C**

Why: Cache for TTL seconds => subsequent lookups skip the hierarchy. TTL is set by the record's owner; high TTL means more cache hits but slower reaction to changes; low TTL means faster updates but more lookup load.

More on DNS: Resource Records and Messages

Q28. [MCQ] The A record

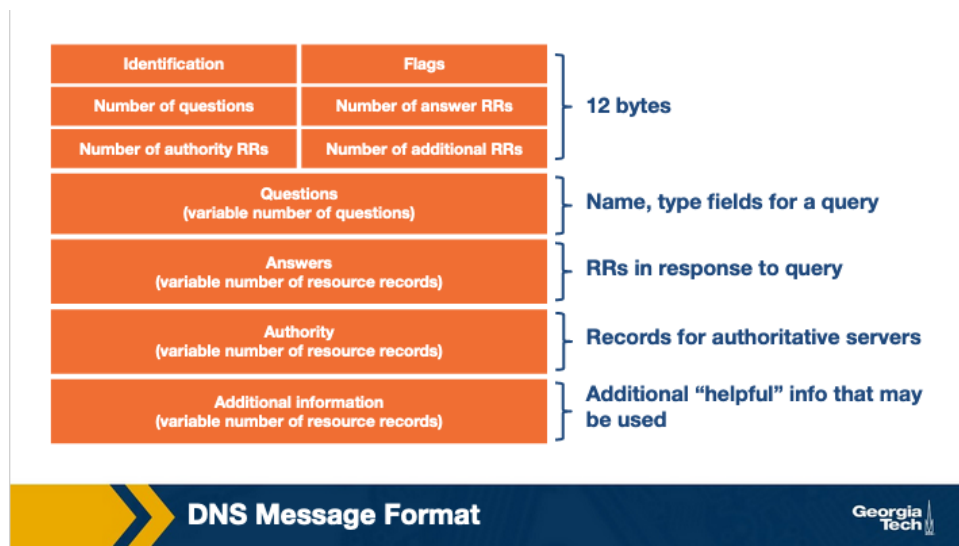


Figure: DNS resource record structure (name, value, type, TTL) (Module 12)

From the resource-record figure, which type maps a hostname to an IPv4 address?

- A. MX
- B. NS
- C. CNAME
- D. A

Correct answer: **D**

Why: A record = hostname -> IPv4 address. The most basic DNS record type; MX is mail exchange, NS is name server, CNAME is alias to another hostname.

Q29. [TF] Resource-record fields

In a DNS resource record (name, value, Type, TTL), the TTL field determines how to interpret the 'value' (e.g., whether it is an IP or a mail server), while the Type field sets the cache lifetime.

- True
- False

Correct answer: **False**

Why: False — the roles are swapped. The TYPE field tells you how to read the value (A=IPv4 address, NS=name server, CNAME=alias, MX=mail server); the TTL sets how long the record may be cached.

Q30. [TF] DNS message sections

In a DNS reply message, the resource records that answer the originally queried hostname appear in the ANSWER section, while the ADDITIONAL section can carry other helpful records (e.g., the IP for a mail server's canonical name).

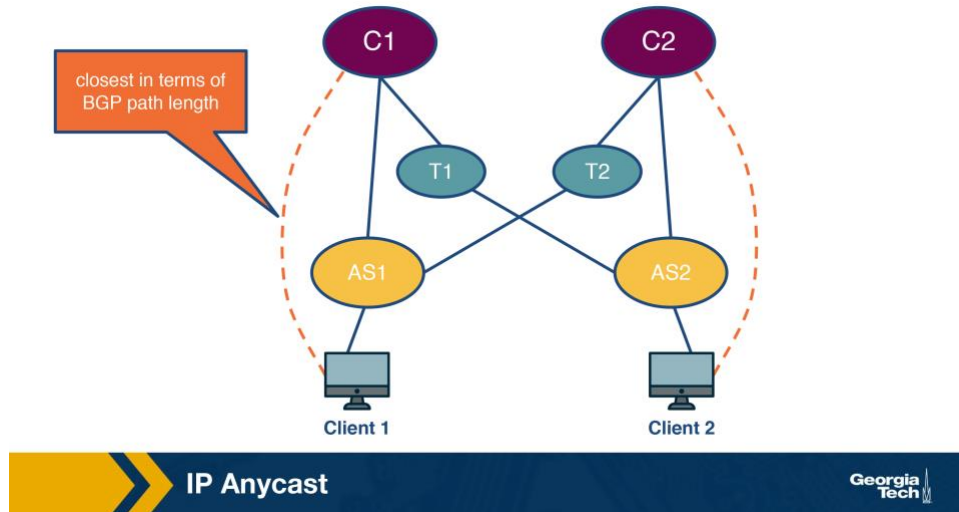
- True
- False

Correct answer: **True**

Why: True. DNS message format: the answer section holds RRs for the queried name; the authority section lists authoritative servers; the additional section carries extra helpful records (e.g., the A record for an MX target).

Server Selection Strategies: IP Anycast

Q31. [MCQ] Anycast routing via BGP



Reference Figure: IP anycast — same IP advertised by multiple clusters (Module 12)

IP anycast gives the same IP address to several replica clusters (C1, C2, ...), and every cluster advertises that one address into BGP — so routers across the Internet see multiple BGP paths that appear to lead to the same destination. In the anycast example, a client in AS1 is routed to cluster C1. Why is the client routed to C1 specifically?

- **A.** C1 advertises the anycast IP and, in BGP terms, the path to C1 is the shortest from AS1; BGP best-path picks the closest replica.
- **B.** C1 has the lowest CPU load at that moment, and the route is computed by an out-of-band CDN controller.
- **C.** C1 was selected by the DNS name server.
- **D.** C1 is the only cluster with the content cached.

Correct answer: **A**

Why: Anycast: same IP advertised from many clusters; BGP picks shortest AS-path. The network itself does the steering — whichever cluster has the shortest BGP path from the client's AS wins routing to that IP.

Q32. [MCQ] Anycast and congestion

Compared with DNS-based redirection, IP anycast for server selection has what main LIMITATION?

- **A.** Anycast requires assigning each replica a UNIQUE IP, which doesn't scale.
- **B.** Anycast picks the 'closest' replica by BGP route, but cannot respond well to dynamic congestion — once BGP installs the route, traffic follows it regardless of moment-to-moment link state.
- **C.** Anycast requires every replica to advertise the prefix from an identical OS image; any version drift between clusters silently corrupts responses to clients.
- **D.** Anycast is the dominant mechanism today for CDN VIDEO delivery.

Correct answer: **B**

Why: Anycast follows BGP routes => can't react to dynamic congestion mid-route. BGP doesn't see link load; once a path is installed, traffic follows it until BGP itself re-converges, which happens on minutes not milliseconds.

Q33. [TF] Anycast: DNS vs TCP flows

IP anycast is ideal for long-lived TCP flows because a mid-connection BGP path change simply continues the same TCP session on whichever replica is now closest.

- True
- False

Correct answer: **False**

Why: False. Anycast suits SHORT, stateless lookups (e.g., DNS). For long-lived TCP flows it is risky: a mid-flow BGP path change can route packets to a DIFFERENT replica that has no state for the connection, breaking it.

Server Selection Strategies: HTTP Redirection

Q34. [MCQ] HTTP redirection overhead

HTTP redirection sends a 3xx response from a 'rendezvous' server pointing the client at a chosen edge. What is the main DOWNSIDE compared to DNS-based selection?

- **A.** HTTP redirection requires a central coordinator to hand out 3xx targets, so it scales worse than DNS, which needs none.
- **B.** HTTP redirection can only move a client within a single cluster, never to a different cluster, so it cannot relieve regional overload.
- **C.** It adds at least one extra RTT (the redirect round-trip) before the client can start fetching content — that latency hurts page-load time.
- **D.** HTTP redirection must resolve the redirected hostname through DNS again, so it always costs two full recursive DNS walks per request.

Correct answer: **C**

Why: HTTP redirect = extra RTT before content fetch. The rendezvous server must reply with 3xx, then the client opens a new TCP to the redirected edge — adding one full round-trip vs DNS-based steering.